



Alberta Public Health Disease Management Guidelines

Brucellosis



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Communicable Disease Control Branch

Public Health Division

Primary and Preventative Health Services

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Table of Contents

Revision History	3
Case Definition	5
Confirmed Case	5
Probable Case	5
Reporting Requirements	6
Physicians, Health Practitioners and Others	6
Laboratories	6
Primary Care Alberta and Indigenous Services Canada	6
Additional Reporting Requirements.....	6
Clinical Assessment and Epidemiology	7
Etiology	7
Clinical Presentation.....	7
Diagnosis	7
Treatment.....	7
Reservoir.....	7
Transmission.....	7
Incubation Period	7
Period of Communicability.....	7
Host Susceptibility	8
Incidence.....	8
Public Health Management.....	9
Key Investigation.....	9
Management of a Case.....	9
Management of Contacts	9
Preventive Measures.....	9
References.....	10

Revision Date	Document Section	Description of Revision
July 2025	General	Entire guideline updated to align with current information
	Case Definition – probable case	Defined epidemiological linkage (footnote).

Brucellosis

Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of *Brucella* species from an appropriate clinical specimen (e.g., blood, bone marrow)^(B);

OR

- Detection of *Brucella* nucleic acid (e.g., PCR) from an appropriate clinical specimen (e.g., blood, CSF, biopsy tissue)^(C);

OR

- Seroconversion or significant (i.e., fourfold or greater) rise in *Brucella* agglutination titre between acute and convalescent phase serum specimens obtained two or more weeks apart.

Probable Case

Clinical illness^(A) and epidemiological-linkage^{(D)(1)};

OR

Clinical illness^(A) and supportive serology (*Brucella* agglutination titre of 1:160 or higher in one or more serum specimens obtained after onset of symptoms).

^(A) Clinical illness: see [Clinical Presentation](#).

^(B) Refer to the [Brucellosis - Public Health \(ProvLab\)](#) for current specimen collection and submission information.

^(C) Refer to the [National Microbiology Laboratory \(NML\) Guide to Services](#) for current specimen collection and submission information.

^(D) Epidemiologically linked includes:

- Direct contact with body fluids or tissue from a confirmed human case of brucellosis, **OR**
- Veterinary occupational exposure to *Brucella* vaccine (i.e., needle stick, mucous membrane exposure), **OR**
- Laboratory exposure to brucellosis-causing *Brucella* species (BBS), **OR**
- Direct contact to an animal diagnosed with a *Brucella* infection (or their fluids), as determined by a veterinarian or the Provincial Health Veterinarian, including potential aerosol exposure, **OR**
- Shared one of the following exposures with a confirmed human case of brucellosis:
 - Consumption of dairy products from a common source that were unpasteurized or of unknown pasteurization, **OR**
 - Consumption or handling of undercooked meat or carcass of an animal from a herd or of a species with a known or suspected history of *Brucella*, **OR**
 - Slaughtering, dressing, butchering, or having other direct contact with animals or animal tissues possibly infected with *Brucella*.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Zone Medical Officer of Health (MOH) (or designate), of all confirmed and probable cases in the prescribed form by mail, fax or electronic submission (e.g., email) within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic submission (e.g., HL7 message) within 48 hours (two business days) to the:

- Zone MOH (or designate), and
- Chief Medical Officer of Health (CMOH) (or designate)

Primary Care Alberta and Indigenous Services Canada

- The Zone MOH (or designate) where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and/or out-of-country cases and/or contacts, the Zone MOH (or designate) should forward the following information to the CMOH (or designate) by phone, fax or electronic submission within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report (if applicable), and
 - other relevant clinical/epidemiological information.

Additional Reporting Requirements

Canadian Food Inspection Agency (CFIA)

Under the [Health of Animals Act – Reportable Diseases Regulations](#), all animal cases of Brucellosis must be reported to CFIA.

If there are known or suspected animal sources of a human brucellosis case, the Zone MOH (or designate) should report them to the District Veterinarian of the CFIA and CMOH (or designate). [CFIA contact information is available here.](#)

Clinical Assessment and Epidemiology

Etiology

Brucellosis is caused by the gram-negative coccobacillus *Brucella*.⁽²⁾ The species known to infect humans are *Brucella abortus* (from cattle), *Brucella melitensis* (from sheep and goats), *Brucella suis* (from swine) and *Brucella canis* (from dogs).^(2,3)

Clinical Presentation

The onset of brucellosis may be acute or insidious with symptoms such as intermittent fever, headache, weakness, profuse sweating, chills, arthralgia, depression, weight loss, and generalized aching.^(2,4) Local infections of organs (e.g., liver, spleen, etc.) may be present.⁽⁴⁾ Serious complications include neurobrucellosis, endocarditis and osteomyelitis.⁽⁵⁾ Brucellosis can occasionally last up to a year or more if not adequately treated.⁽⁴⁾ Untreated cases have a case fatality rate of less than 2%.⁽⁴⁾

Diagnosis

The gold standard of brucellosis diagnosis is confirmation of *Brucella* species from either blood, bone marrow, other tissue specimens by culture.^(2,4) Consultation with the Microbiologist on call is required *prior* to specimen collection. Notify the lab prior to specimen collection and indicate the suspected organism on the requisition. For more information, refer to the [Brucellosis - Public Health \(ProvLab\)](#).

Treatment

Brucellosis is treatable by antimicrobial therapy. Consultation with an infectious disease specialist is recommended.

Reservoir

The most common animal reservoirs are cattle, swine, sheep, and goats.⁽⁴⁾ Other reservoirs may include camels, bison, elk, caribou, moose, wild hogs, dogs and coyotes.⁽⁴⁾

Transmission

Transmission occurs through ingestion (eating or drinking unpasteurized dairy products), direct contact with bodily fluids of infected animals via skin abrasions and mucous membranes, and inhalation.⁽⁴⁾ Airborne transmission has been reported in laboratory and slaughterhouse workers.⁽⁴⁾

Person-to-person transmission is rare.⁽⁴⁾ People who are breastfeeding may transmit infection to their infants.⁽⁴⁾ Although rare, transmission may also occur through blood transfusions, tissue transplants, or sexually.⁽⁴⁾

Brucellosis has the potential to be used as a bioterrorist agent due to its potential to be aerosolized.⁽⁴⁾

Incubation Period

The incubation period is highly variable; generally 1–2 months, with a range of 5 days to 5 months.⁽⁴⁾

Period of Communicability

Unknown. Person-to-person spread is rare.⁽⁴⁾

Host Susceptibility

Susceptibility is universal.

Incidence

Brucellosis occurs worldwide but particularly in the Canadian Arctic, India, Central Asia, Africa, the Mediterranean, the Middle East, Central/South America and Mexico.⁽⁴⁾

Brucellosis has been a nationally notifiable disease in Canada since 1928.⁽¹⁾ For the current incidence in Canada refer to the [Notifiable Diseases Online website](#).

To date, all cases reported in Alberta have been related to travel abroad or residence outside of Alberta (e.g., Canadian Arctic) or Canada. For the current incidence in Alberta, refer to the [Interactive Health Data Application \(IHDA\)](#).

Brucellosis is not endemic in Canadian domestic livestock, and Canada was declared free of bovine brucellosis in 1985 but there are two wildlife reservoirs where the disease persists: free-ranging bison herds in and around Wood Buffalo National Park (located at the border between Alberta and the Northwest Territories), and free-roaming caribou and reindeer in the arctic and sub-arctic regions of Canada.⁽⁶⁾ These reservoirs are managed in order to prevent the spread of brucellosis to domestic livestock, maintaining Canada's brucellosis-free status for livestock.⁽⁶⁾ For more information on brucellosis in livestock and wildlife, refer to the [CFIA website](#).

Public Health Management

Key Investigation

- Confirm that the individual meets the case definition.
- Obtain a history of illness, including date of onset and signs and symptoms.
- Determine possible source of infection, considering the [incubation period](#), reservoir and mode of transmission, and obtain a history of:
 - travel to or immigration from areas where the disease is [endemic](#) in animals (mainly cattle, sheep, and goats),
 - consumption of unpasteurized dairy products or raw meat,
 - occupational risk (e.g., laboratory worker, veterinarian, meat packing plant/slaughterhouse worker, wildlife biologist/officer), and
 - outdoor/lifestyle risk (e.g., hunter).^(4,7)

Management of a Case

- Hospitalized cases should be managed in accordance with hospital infection prevention and control policies.
- Supportive care should be provided based on clinical presentation and dependent on laboratory results.
- Provide general education regarding disease transmission and prevention.

Management of Contacts

- Since brucellosis is rarely transmitted person-to-person, contact follow-up is not generally required.
- However, individuals exposed to the same source (e.g., common exposure to infected animals) should be assessed for clinical illness. Advise them to seek medical evaluation if symptoms develop.

Preventive Measures

- Currently, no human vaccine is available for brucellosis.⁽³⁾
- Educate people on safe food practices, including:
 - Avoid consuming unpasteurized dairy products (e.g., milk, cheese, ice cream), especially when traveling abroad,
 - Refrain from eating uncooked animal viscera, and
 - Cook all domestic and wild meats thoroughly.⁽⁴⁾
- Educate people at risk of exposure (e.g., hunters, animal handlers, farmers, or workers in slaughterhouses, meat processing plants and butcher shops) to:
 - Wear protective gear when handling potentially infected animals, for example:
 - rubber gloves
 - protective clothing
 - face shield and mask.
 - Handle animal carcasses in well-ventilated areas.
 - Dispose of discarded animal remains properly by burying.
 - Exercise caution when handling and disposing of the animal placenta, discharges, and aborted fetuses.
 - Disinfect contaminated areas thoroughly.⁽⁴⁾
- Laboratory workers should adhere to their occupational health and safety protocols.

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